

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTECPE603D - SOFTWARE TESTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Quality Assurance (QA) mainly focuses on:
 - A. Identifying bugs in software
 - B. Ensuring product meets specifications
 - C. Monitoring and improving the process
 - D. Verifying software after release

Which of the following models integrates testing with every phase of development?

- A. Waterfall model
- B. Spiral model
- C. V-Model
- D. Prototype model

- 3 Which of the following best defines "Shift-Left Testing"?
 - A. Testing performed after development
 - B. Testing started early in the development cycle
 - C. Testing delayed to reduce cost
 - D. Testing only during integration

White box testing is also known as:

- A. Behavioral testing
- B. Functional testing
- C. Glass box testing
- D. External testing

Which of the following is not a black box testing technique?

- A. Boundary value analysis
- B. Decision table testing
- C. Statement coverage
- D. Equivalence partitioning

- 6 Which of the following is an advantage of bottom-up integration testing?
 - A. Interface errors can be found early
 - B. Drivers are required
 - C. Stubs are required
 - D. No need for testing low-level modules

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- 7 While choosing an integration method, the most important factor is:
- Developer's experience
 - Module complexity and dependency
 - Programming language used
 - Design style of the database
- 8 What is the purpose of a defect bash?
- Documenting test cases
 - Automating regression testing
 - Finding as many bugs as possible in a short time
 - Performing static code analysis
- 9 Non-functional testing is used to evaluate:
- Application logic
 - Business rules
 - System performance and reliability
 - Syntax correctness
- 10 Acceptance testing is typically performed by:
- Developers
 - Test engineers
 - End users or clients
 - Database administrators
- 11 Regression testing is primarily done to:
- Improve system performance
 - Identify security flaws
 - Ensure new code does not break existing functionality
 - Perform unit testing
- 12 What is defect seeding?
- Removing known bugs before release
 - Adding known defects to check test effectiveness
 - Generating bugs from user input
 - Testing software in production
- Q2. Solve the following.
- Differentiate between Quality, Quality Assurance (QA), and Quality Control (QC) with appropriate examples. 6
 - Describe different phases of Software Testing Life Cycle (STLC)? 6
- Q3. Solve the following.
- Describe the importance of static testing in detecting defects early. Explain the process of code review and walkthrough. 6
 - Describe the process of black box testing. What are the inputs and expected outputs? 6
- Q4. Solve Any Two of the following.
- Describe the process and benefits of bottom-up integration testing. 6
 - Describe use-case-based scenario testing. How are test cases derived from use cases? 6
 - Discuss criteria for selecting an appropriate integration method during software testing? 6
- Q5. Solve Any Two of the following.

- A) Compare functional and non-functional testing with suitable examples. 6
- B) Describe the steps involved in user acceptance testing (UAT). Why is it critical for project success? 6
- C) A customer requires a student portal with features like login, attendance viewing, and grade reports. Write 3–5 acceptance test scenarios that can be used to verify the system before delivery. 6

Q6. Solve Any Two of the following.

- 1. Describe following types of regression testing: 6
 - a) Unit regression b) Partial regression c) Complete regression.
- 2. Suppose you developed a calculator app with multiple features. Describe how you would apply exploratory testing manually to find hidden bugs. 6
- 3. Describe the role of inheritance and polymorphism in object-oriented testing. 6

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